Unit 9 - Computer Networks
Unit 10 - Communication Technologies
Assignment 1 - Upgrade a LAN Tasks 1&2

Introduction

Scenario / Vocational Context - *You are a junior consultant for T5 Alliance. Your new client, Competition Bikes Ltd, have decided that they would like to have wireless capability in their new network. You must adjust your design to include the following new requirements;*

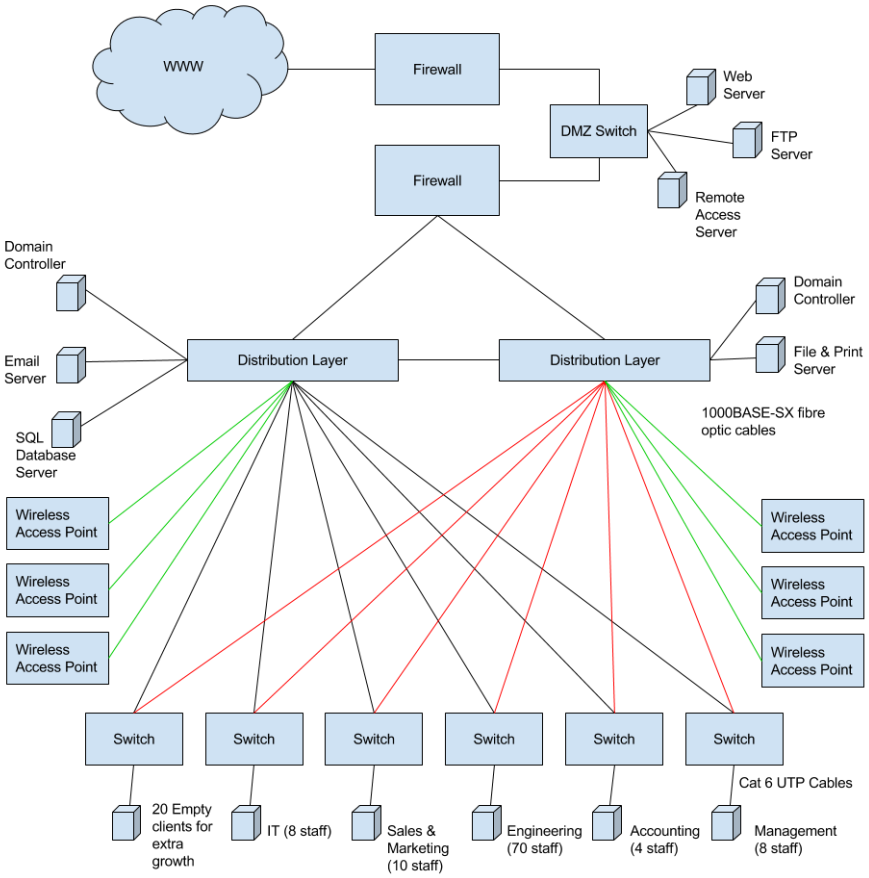
- *Wireless coverage must extend to the whole site.*
- *The wireless network must be as secure as possible.*

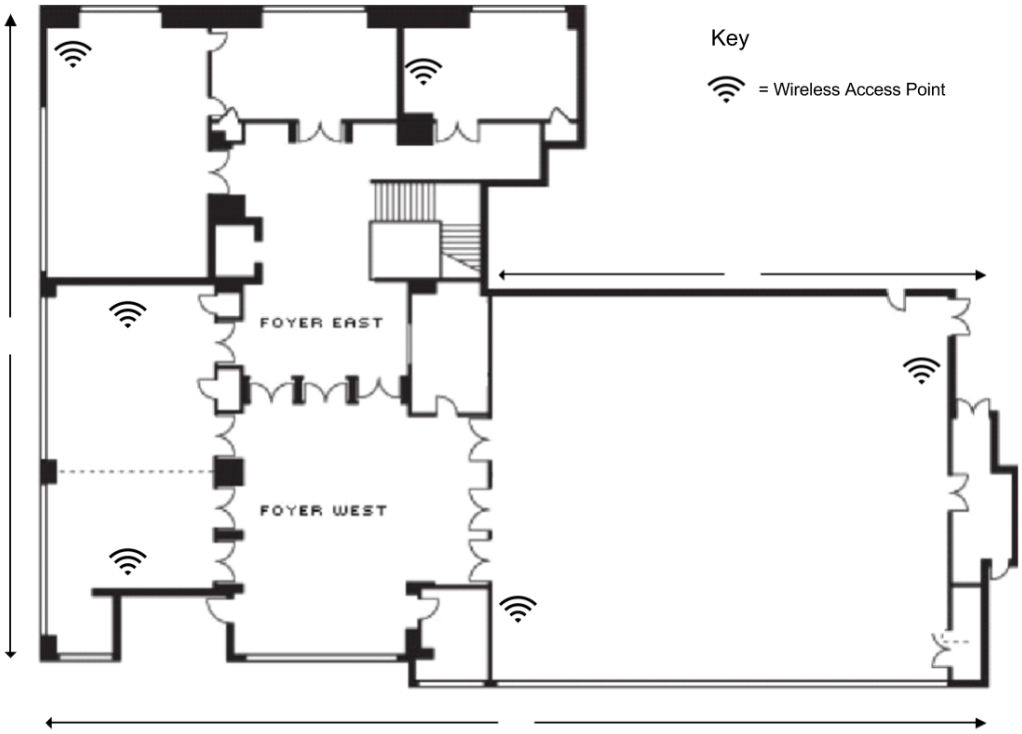
Task 1

(U10P3) - Identify different types of communication devices

Task description - *Upgrade your network topology diagram to include provision for wireless networking, ensuring that your diagram is labelled correctly.*

Upgraded network topology diagram





Task 2

(U10P2) - Identify communication protocols and models / (U9P2) - Describe why different network standards and protocols are necessary

Task description - *Identify at least TWO NEW protocols that have been introduced to your network now that you have included wireless access points. Identify the standards that define the protocols and say why it is necessary for OEMs to use these standards when they design network components.*

Introduction

With the introduction to wireless access points within the network, there are new protocols introduced to the network which have to be identified and managed in order for the network to function to a good standards and with suitable security.

These protocols need to have a standard in which to operate on, meaning that original equipment manufacturer, or OEMs, that manufactures the hardware for wireless connection need to follow these standards in order for their hardware to function properly.

Protocols and standards

802.11n

802.11n is a wireless technology that can include multiple-input multiple-output (MIMO), that has a speed of 450Mbps, which is the 5GHz radio wave frequency. The second transmitter and receiver antennas allow increased data through increased range, which is the 2.4 GHz radio wave frequency, but it is usually slower than the 5GHz radio wave frequency.

To elaborate, the 802.11n protocol uses multiple input and output technology, which is known as MIMO. This is so it has a wider radio frequency channel, giving it substantial range.

Essentially, 802.11n is a specification for the wireless LAN communication standards, those standards falling under the IEEE 802.11 family. Which is a set of media access control protocols. The IEEE 802.11 protocol is one way of accessing a network topology.

It allows a wireless device or technology to connect to a wired networking infrastructure. By allowing this protocol to work a mobile phone user can now walk around the rooms within the network and still have a strong signal to the Wi-Fi. However, It is not just places of business that have this standard, colleges, airports and other public places can provide this.

IEEE is short for Institute of Electrical and Electronic Engineers and it is their responsibility to judge how different wireless devices are made and how they communicate with one another. The letters which follow “802.11” mean a change has been made so 802.11n is much newer than a 802.11g network.

To elaborate, 802.11 And 802.11x is a family of wireless technologies that have been developed by the IEEE. 802.11 is an over the air (OTA) interface that is between a wireless client e.g. a smartphone and a base station.

It can also be used between two wireless clients. E.g. an iPhone can be tethered with an iPad to provide Wi-Fi between the two. There are quite a few different specifications in the 802.11 family.

802.11 applies to wireless Local Access Networks and will provide a 1-2 mbps speed the 2.4GHz frequency. 802.11a is an extension to 802.11 that applies to wireless Local Access Networks and provides up to 54-Mbps on the 5GHz frequency. 802.11b is an extension that can provide 11 Mbps speeds on the 2.4GHz frequency.

802.11g applies to wireless LANs and used for wireless data transmission over short distances at up 54 Mbps in the 2.4GHz frequency. 802.11n is a wireless technology that can include multiple-input multiple-output (MIMO). 802.11ac can deliver data speeds of up to 433Mbps per spatial stream or 1.3Gbps in a three antenna design. This operates on the 5GHz frequency.

Newer versions of this standard often improve the performance and data rate. Data rate is the speed and how fast data can travel over the wireless network. The range is also another improvement which is obviously the distance the signals can reach.

These standards within a wireless device that can send out radio signals is important to have, due to OEMs usually making their products different to other manufacturer. So, having standards implemented for every OEM means that all the products manufactured can be connected to each other, no matter the manufacturer.

Essentially, communication protocols are almost like someone who is directing the traffic on a road, they are almost a traffic guard who are controlling the traffic. Essentially, communication protocols will organise everything when the computers are communicating with each other.

Basically, without communication protocols all the signals would get mixed and the computers wouldn't be able to communicate. So, when all the computers are connected they can find the location of the computer and can transfer information to one another.

Overall, the reason that it is important to have this standard is so that communication Protocols are a set of rules. Without them it would be chaos, causing packets to be lost because there would be so many collisions, or the devices just wouldn't communicate with each other, making the wireless network obsolete.

WPA2 / 802.11i

WPA2, or Wi-Fi Protected Access II, is a security protocol which is commonly found with modern wireless computer networks. Basically, this is where you put in your password to get onto the Wi-Fi network.

WPA2 falls under the 802.11i protocol standard, which is basically a standard for wireless local area networks. It provides an improved encryption for networks that use the 802.11a, 802.11b and 802.11g, which are very popular. However, it can be used for other 802.11 protocols but are very rare.

There are many different types of wireless security protocols, some of which fall under the 802.11i security protocol standard. The three main types are WEP, WPA AND WPA2, with only WPA and WPA2 falling under the 802.11i security standard.

The original protocol is called Wired Equivalent Privacy (WEP), it was designed for encrypting wireless networks. It was made for wireless networks and is called a wired protocol because it was created to give the same security as a wired network.

WPA is short for Wi-Fi Protected Access and was introduced when the 802.11i security was being made, which was somewhere around 2004. WPA Enterprise uses an Authentication Server, which will generate certificates and keys.

The second version (WPA2) is an improvement over the first one. It uses an Advanced Encryption Standard to encrypt the files. WPA2 is very secure, so secure that the government even use it to protect files from any hackers.

Wireless security protocols are designed to encrypt the private data on your computer as it is being sent to another computer so that your data is more secure. Essentially, this type of security protocol creates an encrypted code which keep everything the data that is being sent and received secure from unauthorised access.

Without security protocols, not only the wireless network, but the entire network may be infiltrated by an unauthorised access and the data within the network might be at risk of being removed or stolen.

If the wireless network had a proper security protocol installed, the data stored within the wireless network and the rest of the network will be at less risk of being attacked by an outside source.

Overall, having standard security protocols as a set of rules for the manufacturers is very important to have, without them it would be catastrophic because there would such a high risk of the data being taken or removed, causing a lot of problems within the network.

Conclusion

In conclusion, it is very important for original equipment manufacturer to stick with the standards of wireless networking. The reason being that these standards are placed there so the equipment produced by the manufacturer is compatible with other manufacturers which build similar equipment, but design them differently to others.

Essentially, these standards exist so a manufacturer doesn't decide to create a different method of security, radio wave protocol or data transferring, even if they think it's better for their product. This is so the equipment isn't obsolete to other forms of equipment built outside the company, and that other devices need less software and hardware restrictions in order to connect to wireless networks .